4.3 Ablation Studies

In this section, we perform ablation studies to study the effects of various hyper parameters used in our model. We perform the ablation studies on DenseNet-BC [8] network with CIFAR-100 [10] as in-distribution while training and iSUN [20] as the OOD validation data while testing. By default, we use 5 random splits for CIFAR-100, $\epsilon = 0.002$, *SFX+MarginEntropy* loss to train network, accuracy bound $\delta = 2\%$ to save the models, and use *(Softmax + Entropy)@Temperature* with $T = 1000$ to detect out-of-distribution samples. Results are given in Table 2.

(1) **Number of splits:** This analysis characterizes the sensitivity of our algorithm to the number of splits of the training classes which is same as the number of classifiers in the ensemble. As the number of splits increase, the number of times a particular training class being in-distribution for the leave-out classifiers increases too. This enables the ensemble to discriminate an in-distribution sample from the OOD sample. But it also increases the computational cost. For CIFAR-100, we studied 3, 5, 10 and 20 splits. Our results show that while 5 splits gave the best result, 3 splits also provides a good trade-off between accuracy and computational cost. We choose the number of splits as 5 as default value.

(2) Type of splits: This study characterizes the way in which the classes are split into mutually exclusive sets. We experiment with splitting the classes manually using prior knowledge and splitting randomly. For the manual split, the class labels are first clustered into semantically consistent groups and classes from each group are then distributed across the splits. The results show that the OOD detection rates for random selections are better than the manual selection.This ensures that we can achieve good OOD detection rates even by random selection of classes when the number of classes is huge.

(3) Different ϵ for input preprocessing: For input preprocessing, we sweep over $\epsilon \in [0, 0.000313, 0.000625, 0.00125, 0.002, 0.003]$. Our results show that as ϵ increases from 0, the performance of out-of-distribution detector increases, and it reaches the best performance at 0.002. The further increase of ϵ does not help performance.

(4) Different T for temperature scaling: For temperature scaling, we sweep over $T \in [1, 10, 100, 1000, 5000]$. Our results show that for DenseNet-BC with CIFAR-100, as T increases from 1 to 1000, the performance of out-of-distribution detector increases. Beyond $T = 1000$, the performance does not change much.

(5) Loss function variants: We study the effects of training our method with different types of losses. The training regime follows the strategy given in section 3.2, where the training data X is split into K partitions $X_i, i \in \{1, ..., K\}$. A total of K classifiers are trained where classifier F_i uses the partition X_i as OOD data and $X - X_i$ as in-distribution data.

- *SFX*: We assign an additional label to all the OOD samples and train classification network using the cross entropy loss.

- *SFX+MaxEntropyDiff*: Along with the cross entropy loss, we maximize the difference between the entropy of in- and out-of-distribution samples across all in-distribution classes.
- *SFX+MarginEntropy*: Along with the cross entropy, we maximize the difference between the entropy of in- and out-of-distribution samples across all in-distribution classes, but is bounded by a margin as given in Equation 1.

Our results show that the proposed *SFX+MarginEntropy* loss works dramatically better than all other types of losses for detecting out-of-distribution samples as well as for accurate classification. The results demonstrate that the proposed novel loss function *SFX+MarginEntropy* (equation 1) is the major factor in significant improvements over the current state-of-the-art ODIN [13].

(6) Out-of-distribution detector: We study different OOD scoring methods to discriminate out-of-distribution samples from in-distribution samples.

- *Softmax score*: Given an input image, the score is given by the average of maximum softmax outputs over all the classifiers in the ensemble.
- *Entropy score*: Given an input image, the score is given by the average of entropy of softmax vector over all the classifiers in the ensemble.
- *Softmax + Entropy*: Given an input image, both the above scores are added.
- *Softmax@Temperature*: Given an input image, the above described *S*oftmax score is computed on temperature scaled ($T = 1000$) softmax vectors.
- *Entropy@Temperature*: Given an input image, the above described *E*ntropy score is computed on temperature scaled ($T = 1000$) softmax vectors.
- *(Softmax + Entropy)@Temperature*: Given an input image, the above described *E*ntropy@Temperature and *S*oftmax@Temperature are computed ($T = 1000$) on softmax vectors and then added.

Among the above OOD scoring methods, the *(Softmax + Entropy)@Temperature* ($T = 1000$) achieved the best performance. *Softmax@Temperature* ($T = 1000$) achieved the second best performance.

4.4 Results and Analysis

Table 3 shows the comparison between our results and ODIN [13] on various benchmarks. The results are reported on all neural network, in-dataset and OOD dataset combinations. Our hyperparameters are tuned using iSUN dataset. From the Table 3, it is very clear that our approach significantly outperforms ODIN [13] across all neural network architectures on almost all of the dataset pairs. The combination of novel loss function, OOD scoring method, and the ensemble of models has enabled our method to significantly improve the performance of OOD detection on more challenging datasets, such as LSUN (resized), iSUN and ImageNet(resized), where the images contain full objects as opposed to the cropped parts of objects. The proposed method is slightly worse on the uniform and some of Gaussian distribution results. Moreover, our method achieves significant gains on both CIFAR-10 and CIFAR-100 with the same number of splits which is 5, even though the number of classes have increased by a factor of

	OOD Dataset	FPR at 95% TPR ↓	Detection Error ↓	AUROC ↑	AUPR In ↑	AUPR Out ↑
		each cell in **ODIN[13]/Our Method** format				
DenseNet-BC CIFAR-10	TINc	4.30/**1.23**	4.70/**2.63**	99.10/**99.65**	99.10/**99.68**	99.10/**99.64**
	TINr	7.50/**2.93**	6.10/**3.84**	98.50/**99.34**	98.60/**99.37**	98.50/**99.32**
	LSUNc	8.70/**3.42**	6.00/**4.12**	98.20/**99.25**	98.50/**99.29**	97.80/**99.24**
	LSUNr	3.80/**0.77**	4.40/**2.1**	99.20/**99.75**	99.30/**99.77**	99.20/**99.73**
	UNFM	**0.00**/2.61	**0.20**/3.6	**100**/98.55	**100**/98.94	**100**/97.52
	GSSN	**0.00**/**0.00**	**0.50**/0.2	**99.90**/99.84	**100**/99.89	**99.90**/99.6
DenseNet-BC CIFAR-100	TINc	17.30/**8.29**	8.80/**6.27**	97.10/**98.43**	97.40/**98.58**	96.80/**98.3**
	TINr	44.30/**20.52**	17.50/**9.98**	90.70/**96.27**	91.40/**96.66**	90.10/**95.82**
	LSUNc	17.60/**14.69**	9.40/**8.46**	96.80/**97.37**	97.10/**97.62**	96.50/**97.18**
	LSUNr	44.00/**16.23**	16.80/**8.77**	91.50/**97.03**	92.40/**97.37**	90.60/**96.6**
	UNFM	**0.50**/79.73	**2.50**/9.46	**99.50**/92.0	**99.60**/94.77	**99.00**/83.81
	GSSN	**0.20**/38.52	**1.90**/8.21	**99.60**/94.89	**99.70**/96.36	**99.10**/90.01
WRN-28-10 CIFAR-10	TINc	23.40/**0.82**	11.60/**2.24**	94.20/**99.75**	92.80/**99.77**	94.70/**99.75**
	TINr	25.50/**2.94**	13.40/**3.83**	92.10/**99.36**	89.00/**99.4**	93.60/**99.36**
	LSUNc	21.80/**1.93**	9.80/**3.24**	95.90/**99.55**	95.80/**99.57**	95.50/**99.55**
	LSUNr	17.60/**0.88**	9.70/**2.52**	95.40/**99.7**	93.80/**99.72**	96.10/**99.68**
	UNFM	**0.00**/16.39	**0.20**/5.39	**100**/96.77	**100**/97.78	**100**/94.18
	GSSN	**0.00**/**0.00**	**0.10**/1.03	**100**/99.58	**100**/99.71	**100**/99.2
WRN-28-10 CIFAR-100	TINc	43.90/**9.17**	17.20/**6.67**	90.80/**98.22**	91.40/**98.39**	90.00/**98.07**
	TINr	55.90/**24.53**	23.30/**11.64**	84.00/**95.18**	82.80/**95.5**	84.40/**94.78**
	LSUNc	39.60/**14.22**	15.60/**8.2**	92.00/**97.38**	92.40/**97.62**	91.60/**97.16**
	LSUNr	56.50/**16.53**	21.70/**9.14**	86.00/**96.77**	86.20/**97.03**	84.90/**96.41**
	UNFM	**0.10**/99.9	**2.20**/14.86	**99.10**/83.44	**99.40**/89.43	**97.50**/71.2
	GSSN	**1.00**/98.26	**2.90**/16.88	**98.50**/93.04	**99.10**/88.64	**95.90**/71.62

Table 3: Distinguishing in- and out-of-distribution test set data for the image classification. All values are percentages. ↑ indicates larger value is better, and ↓ indicates lower value is better. Each value cell is in "ODIN[13]/Our Method" format.

ten from CIFAR-10 to CIFAR-100. Thus the number of splits need not be scaled linearly with the number of classes, making our method practical. We implicitly outperform Hendrycks and Gimpel [7] and Lee *et.al,* [12] as ODIN outperforms both these works and our method outperform ODIN on all but two benchmarks.

All three components in our method, namely novel loss function, the ensemble of leave-out classifiers and improved OOD detection metric contributed to the improvement in performance over state-of-the-art ODIN (refer to table 2). The contribution of these components can be seen in Table 2 in the rows marked as "Loss function", "Number of splits" and "OOD detection scores".

Our algorithm has stochasticity in the form of random splits of the classes. Given 100 classes in CIFAR-100, there are many ways to split 100 classes into 5 partitions. Table 4 gives the mean and standard deviation across five random ways to partition data when we use 5 number of splits for training. We note that even our worst case results outperform ODIN [13] on more challenging datasets.

Figure 1 compares the histogram of OOD detection scores on ID and OOD samples when different loss functions are used for training. Figure 1(a) is trained